SEMINAR SUMMARY: MARCH 24, 2026 - REED OGROSKY SOME MATHEMATICAL MODELING PROJECTS USING DIFFERENTIAL EQUATIONS IN FLUID DYNAMICS

ROB TOMPKINS

1. Mathematical modeling of gas—liquid transport in human airways/lungs

The lecture began with a presentation of a model of the fluid mechanics of the movement of mucus around the lungs, which is a curious problem because there is quite a large free surface of liquid in the lungs due to the branches as well as the fluid itself being non-newtonian (having viscoelastic properties) in nature. We began by discussing the mechanics of the cilia that line the lungs and how they beat in rhythmic fashions to move the mucus up and out of the lungs generally. We were then posed with a quote:

> "It is awkward that in 2022 a comprehensive description of the organization and topography of mucus transport macroscopically under basal or stressed conditions is not available." — Hill et al. (2022)

Date: March 26, 2026.